2025

COMPUTER SCIENCE — GENERAL

Paper : DSE-B-3

(Computational Mathematics)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question no.*** **1** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) Differentiate between Weighted graphs and unweighted graphs.

(b) What is the difference between Newton Forward and Newton Backward Interpolation?

(c) State any two properties of a coefficient matrix in a system of linear equations.

(d) Explain, why Gauss Elimination can fail for certain systems of equations?

(e) Write the iterative formula used in the Newton-Raphson method.

(f) What is the main idea behind the Bisection method?

(g) Write down the formula for Simpson's 1/3 rd rule.

(h) What is a planar graph? Give an example.

2. (a) Define and differentiate between absolute error, relative error, and percentage error.

(b) Derive the formula for Newton Forward Interpolation and apply it to find $f(x)$ at $x = 1.5$ given the following table :

x	$f(x)$
1	2.0
2	4.4
3	7.9

4+6

3. (a) What do you mean by Linear Dependency of Vectors?

(b) Solve the following system of equations using Gauss Jordan Elimination :

$$\begin{aligned} x + y + z &= 6 \\ 2x + 3y + 7z &= 20 \\ x + 3y + 4z &= 13 \end{aligned}$$

3+7

Please Turn Over

4. (a) Write the composite expression for Trapezoidal rule.

Evaluate $\int_1^3 \frac{dx}{1-x^2}$, by using Trapezoidal rule. 5+5

(b) Solve $f(x) = \cos(x) - x = 0$, using Bisection method correct upto two decimal places.

5. (a) Solve $e^x - 4x = 0$, using the Newton-Raphson method (correct to 4 decimal places).

(b) Discuss on graph coloring problem. What is the chromatic number of a bipartite graph? 5+(3+2)

6. (a) Prove that the number of odd degree vertices in a graph is always even.

(b) When two graphs are said to be isomorphic? Discuss with suitable example.

(c) Prove that the total degree of a graph is always even. 4+3+3

7. (a) What is Euler graph? Show that a tree with n vertices has exactly (n – 1) edges.

(b) When a vertex is said to be isolated? Write the degree of the vertex.

(c) Define path and circuit. What is regular graph? Give an example. (2+3)+2+3

8. (a) Given the table :

x	0	1	2	3
$f(x)$	1	2	9	28

Extimate $f(1.5)$ using Newton's backward interpolation.

(b) Evaluate $\int_0^1 \sqrt{(1+x^3)}\, dx$ using Composite Simpson's rule ($h = 0.25$). 5+5